One Unit of Existence

The Lifecycle of a Hydrogen Atom
from Lattice Geometry to Dissolution

A Formal Treatment in Foundational Ternary Dynamics

Abstract

We present the lifecycle of a hydrogen atom as a deductive chain within Foundational Ternary Dynamics (FTD), a discrete computational framework postulating a three-dimensional cubic lattice with ternary states and local causality. The central mathematical object is the *master quadratic* $x^2 - 16G^{*2}x + 16G^{*3} = 0$, whose coefficients are determined by the lemniscatic constant $G^* = \sqrt{2}\,\Gamma(1/4)^2/(2\pi)$. Its two roots yield $x_+ = 137.036\ldots$ (identified with α^{-1} at 1.26 ppm) and $\lfloor x_- \rfloor = 3$ (the number of color charges, uniquely self-consistent in $D = 3$). From this single algebraic object, the gauge groups $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, the Higgs quartic $\lambda = 3/23$, the Bell violation $S = 2\sqrt{2}$, and the full hydrogen spectrum follow—each stage of the atom's existence governed by the same constant in different clothing. Every claim is tagged with its epistemic status: axiom, theorem, selection, or conjecture. The five selection principles on which the strongest results depend are stated and critiqued. Falsification criteria are provided.

"What is a hydrogen atom?" is a question that admits an answer at every level of description. This is the answer at the level where the description itself becomes the thing described.

1 The Axiom

Before there is a hydrogen atom, before particles, before space itself—what is the minimal ontology from which physics can be built?

Axiom 1 (Axiom Zero)**.** [Axiom] Reality consists of *voxels.* Each voxel has exactly two irreducible properties:

1. **State:** $s \in \{-1, 0, +1\}$ (ternary).
2. **Position:** $x \in \mathbb{Z}^3$ (cubic lattice).

From this single axiom, the remaining postulates of FTD follow as consequences or minimal extensions:

Definition 2 (Two-Layer Ontology)**.** [Axiom] The framework posits a graded ontology:

Layer	**Object**	**Status**
Flux field	$\mathbf{J}(v) \in \mathbb{R}^3$ at each $v \in \mathbb{Z}^3$	Dispositional (tendency)
State field	$s(v) \in \{-1, 0, +1\}$	Actual (manifested)

The flux field encodes what the substrate *would do* if conditions for manifestation were met. The state field encodes what *is*.

Axiom 3 (Local Causality)**.** [Axiom] Information propagates at most one lattice unit per discrete time step (tick). The causal neighborhood is the 26-site Moore neighborhood of $\mathbb{Z}^3$.

Axiom 4 (Determinism)**.** [Axiom] The update rule $\big(s(v,t),\, \mathbf{J}(v,t)\big) \mapsto \big(s(v,t+1),\, \mathbf{J}(v,t+1)\big)$ is a deterministic function of the Moore neighborhood state at time t.

Remark 5. The void is not emptiness. It is the cancellation $0 = (-1)+(+1)$: the maximally symmetric state in which every tendency is balanced by its complement. In Platonic terms, it is the Receptacle (*Timaeus* 48e–52d): the formless substrate that receives determination from the Forms.

2 The Geometry

The lattice exists. What is its intrinsic geometry? The answer lies in a curve that predates the lattice by two centuries: the lemniscate of Bernoulli, $r^2 = \cos 2\theta$.

Definition 6 (The Quartic Integral)**.** [Theorem]

$$I_4 \;=\; \int_0^1 \frac{dx}{\sqrt{1-x^4}} \;=\; \frac{\Gamma(1/4)^2}{4\sqrt{2\pi}} \;=\; 1.31103\ldots$$

This requires only the integer 4 as input.

Theorem 7 (Beta Function Evaluation)**.** [Theorem] *Substituting* $u = x^4$ *gives* $I_4 = \frac{1}{4}\,B(1/4,\,1/2)$*. The reflection formula* $\Gamma(1/4)\,\Gamma(3/4) = \pi\sqrt{2}$ *and* $\Gamma(1/2) = \sqrt{\pi}$ *yield* $I_4 = \Gamma(1/4)^2/(4\sqrt{2\pi})$*.*

Definition 8 (The Lemniscate Constant and the Bridge Constant)**.** [Theorem]

$$\varpi \;=\; 2\,I_4 \;=\; 2.62206\ldots \qquad\qquad G^* \;=\; \frac{2\varpi}{\sqrt{\pi}} \;=\; \frac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi} \;=\; 2.95868\ldots$$

The constant ϖ is the half-perimeter of the lemniscate. The constant G^* absorbs the geometric scaling between the lemniscatic and circular domains, yielding the identity

$$\pi \;=\; \frac{4\,\varpi^2}{G^{*2}}\,. \tag{1}$$

Remark 9. In FTD, the number π is not fundamental. It is 4λ, where $\lambda = \varpi^2/G^{*2} = \pi/4$ is the *ontic ratio*: the bridge between the self-intersecting (lemniscatic) and the circular (Euclidean) geometries.

The bridge constant G^* is not arbitrary. It is an invariant of a specific algebraic object.

Definition 10 (The Elliptic Curve)**.** [Theorem] Let $E : y^2 = x^3 - x$ over $\mathbb{Q}$. This curve has j-invariant $j(E) = 1728 = 12^3$, endomorphism ring $\mathrm{End}(E) \cong \mathbb{Z}[i]$ (Gaussian integers), and automorphism group $|\,\mathrm{Aut}(E)| = 4$.

Selection Principle 11 (SP1: The CM Curve). [Selection] Among all elliptic curves, we select $E : y^2 = x^3 - x$ with $j = 1728$. **Justification:** it has the maximal automorphism group ($\mathbb{Z}_4$) compatible with square lattice symmetry, since $\mathbb{Z}[i] \cong \mathbb{Z}^2$ as additive groups. **Critique:** The curve with $j = 0$ has $|\,\mathrm{Aut}\,| = 6 > 4$, but hexagonal, not square, symmetry. No proof excludes non-CM curves.

Theorem 12 (Watson Integral Connection)**.** [THEOREM] *The Watson integral of the three-dimensional cubic lattice is*

$$W_3 \;=\; \frac{1}{\pi^3}\int_0^\pi\!\int_0^\pi\!\int_0^\pi \frac{d\theta_1\, d\theta_2\, d\theta_3}{3-\cos\theta_1-\cos\theta_2-\cos\theta_3} \;=\; \frac{\Gamma(1/4)^4}{4\pi^3} \;=\; 1.3932\ldots$$

and satisfies $G^* = \sqrt{2\pi\, W_3}$*. This identity connects the lattice Green's function of* $\mathbb{Z}^3$ *to the geometry of the lemniscatic elliptic curve.*

3 The Master Quadratic

The geometry has produced a constant. The constant now produces an equation.

Selection Principle 13 (SP2: The Quadratic Form). [SELECTION] The master equation is a polynomial of degree 2 in x with coefficients determined by G^*. **Justification:** a quadratic is the minimal polynomial admitting two distinct roots; Vieta's formulas express both roots through simple powers of G^*. **Critique:** no principled argument excludes cubic or transcendental forms.

Definition 14 (The Master Quadratic)**.** [SELECTION]

$$x^2 \;-\; 16\, G^{*2}\, x \;+\; 16\, G^{*3} \;=\; 0\,. \tag{2}$$

The coefficient $16 = |\,\mathrm{Aut}(E)|^2 = 4^2$ is an intrinsic arithmetic invariant of E, appearing through six independent routes: the endomorphism ring units, the torsion group $|E(\mathbb{Q})_{\mathrm{tors}}|^2$, the conductor $N/2 = 32/2$, the discriminant $|\Delta|/4 = 64/4$, the BSD denominator, and the unique non-trivial Lucas perfect square $L_3^2 = 4^2$.

Theorem 15 (The Two Roots)**.** [THEOREM] *The discriminant* $\Delta = 256\, G^{*4} - 64\, G^{*3} = 64\, G^{*3}(4G^* - 1) > 0$ *since* $G^* > 1$*. Both roots are positive real:*

$$x_+ = 8\, G^{*2}\left(1 + \sqrt{1 - \tfrac{1}{4G^*}}\right) = 137.036\,171\ldots \tag{3}$$

$$x_- = 8\, G^{*2}\left(1 - \sqrt{1 - \tfrac{1}{4G^*}}\right) = 3.023\,964\ldots \tag{4}$$

Proof. Direct application of the quadratic formula to (2). The product $x_+ x_- = 16\, G^{*3} > 0$ and sum $x_+ + x_- = 16\, G^{*2} > 0$ confirm both roots are positive. ☐

Conjecture 16 (Physical Identification)**.** [CONJECTURE] *The roots correspond to physical constants:*

1. $x_+ = \alpha^{-1}$*, the inverse fine-structure constant. The CODATA 2022 value* $\alpha^{-1} = 137.035\,999\,177(21)$ *agrees at 1.26 ppm.*
2. $\lfloor x_- \rfloor = N_c = 3$*, the number of color charges in QCD.*

Remark 17. Conjecture 16 is the framework's sharpest empirical contact point *and* its most vulnerable. If x_+ diverges from α^{-1} beyond the precision formula's reach, the entire framework collapses (see §11).

4 Three Dimensions, Three Colors

The universe cools. At roughly 10^{-5} s after whatever initiated the expansion, the temperature drops below $\sim 150\,\mathrm{MeV}$—the QCD confinement scale. Three quarks (two up, one down) are confined into a proton: a discrete, countable object (baryon number = 1) whose internal structure is a continuous quantum field theory. The proton is made of motion, not of matter.

The number of colors is not arbitrary. It is selected by the spatial dimension.

Theorem 18 ($D = 3$ Uniqueness)**.** [THEOREM] *Let W_D be the Watson integral of $\mathbb{Z}^D$, $G^*_D = \sqrt{2\pi\, W_D}$, and $x_\pm^{(D)}$ the roots of $x^2 - 16\, {G^*_D}^2\, x + 16\, {G^*_D}^3 = 0$. Then $\lfloor x_-^{(D)} \rfloor = D$ if and only if $D = 3$.*

D	W_D	G^*_D	$x_+^{(D)}$	$x_-^{(D)}$	$\lfloor x_-^{(D)} \rfloor \stackrel{?}{=} D$
1	∞	—	—	—	N/A
2	11.99	8.678	1196	8.74	**No** ($8 \neq 2$)
3	1.393	2.959	137.0	3.024	**Yes** ($3 = 3$)
4	1.118	2.651	109.7	2.716	**No** ($2 \neq 4$)
5	1.047	2.565	102.6	2.631	**No** ($2 \neq 5$)
6	1.020	2.532	100.0	2.598	**No** ($2 \neq 6$)

Scholium 19. This is a self-referential selection: the dimension determines the Watson integral, which determines G^*, which determines the gap equation, whose smaller root yields $N_c = \lfloor x_- \rfloor$. Only $D = 3$ closes the loop $N_c = D$. Three spatial dimensions are not an axiom; they are the unique self-consistent solution.

Theorem 20 (Gauge Groups from Moore Decomposition)**.** [THEOREM] *The 26 neighbors of a voxel in $\mathbb{Z}^3$ decompose uniquely by distance into three sublattices:*

$$26 \;=\; \underbrace{6}_{SC} + \underbrace{12}_{FCC} + \underbrace{8}_{BCC}\,.$$

A neighbor at displacement $(\Delta x, \Delta y, \Delta z)$ couples to the $\mathbf{J}$-components corresponding to its nonzero coordinates. The number of excited components determines the gauge symmetry:

Sublattice	*Count*	*Distance*	$\mathbf{J}$*-components*	*Gauge group*
SC	*6*	*1*	*1*	U(1)
FCC	*12*	$\sqrt{2}$	*2*	SU(2)
BCC	*8*	$\sqrt{3}$	*3*	SU(3)

The Standard Model gauge group $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ *is the unique orthogonal decomposition of $|\mathbf{J}|^2 = J_x^2 + J_y^2 + J_z^2$ by the Moore sublattices.*

Remark 21. The one-loop QCD beta function coefficient $b_3 = (11N_c - 2N_f)/3 = 7$ (for $N_f = 6$ quark flavors) sets the running of the strong coupling. The proton's confinement is controlled by the same quadratic that will later determine its electromagnetic binding to an electron.

5 The Electron and the Atom

For the first 380,000 years, the universe is a plasma. Then the temperature drops below $\sim 3000\,\mathrm{K}$, the photon bath no longer ionizes hydrogen, and nearly every proton captures an electron. The universe becomes transparent. This is the moment our hydrogen atom is born.

Definition 22 (Manifestation Threshold)**.** [Selection] A voxel transitions from $s = 0$ to $s = \pm 1$ when the local flux density exceeds the threshold $K_B = 0.511\,\mathrm{MeV}/c^2$. This identifies K_B with the electron mass m_e.

Proposition 23 (Electron Mass)**.** [Selection]

$$m_e \;=\; m_P\,\sqrt{2\pi}\,\frac{16}{3}\,\alpha^{11} \;=\; 0.5100\ \mathrm{MeV}/c^2$$

where $m_P = 1.2209 \times 10^{19}$ *GeV is the Planck mass. The power* α^{11} *encodes the hierarchy:* α^8 *(electromagnetic suppression from Planck to electroweak) times* α^3 *(Yukawa structure). Accuracy:* 0.20% *vs. PDG* 0.51100 *MeV*$/c^2$*.*

Proposition 24 (Higgs Quartic from Ternary Decomposition)**.** [Selection] *The three ternary states* $\{-1, 0, +1\}$ *decompose as 2 active (manifested) + 1 void (unmanifested). Identifying the gauge weights* $w_{\mathrm{SU}(2)} = 2$*,* $w_{\mathrm{U}(1)} = 1$ [Selection] *yields:*

$$\lambda \;=\; \frac{\sin^2\theta_W}{2 - \sin^2\theta_W} \;=\; \frac{3/13}{2 - 3/13} \;=\; \frac{3}{23} \;=\; 0.1304\ldots$$

where $\sin^2\theta_W = 3/13$ *from the ternary Weinberg angle. The Higgs vacuum expectation value* $v = m_P\sqrt{2\pi}\,\alpha^8 = 246.08$ *GeV* [Selection] *gives:*

$$m_H \;=\; v\sqrt{\frac{6}{23}} \;=\; 125.69\ \mathrm{GeV} \quad (0.35\%\ \mathrm{vs.\ PDG}\ 125.25\ \mathrm{GeV}).$$

The algebra is rigorous given the gauge weight identification; the identification itself is the selection.

The hydrogen atom now exists. Its entire energy spectrum follows from a single root of the master quadratic.

Proposition 25 (Hydrogen Spectrum)**.** [Parametric Insertion] *The energy levels of hydrogen are given by standard quantum mechanics:*

$$E_n \;=\; -\frac{\alpha^2\,m_e c^2}{2n^2}\,, \qquad n = 1, 2, 3, \ldots$$

FTD provides the inputs (α, m_e)*; the formula is external. Using the FTD-derived values* $\alpha^{-1} = 137.036$ *and* $m_e = 0.5100$ *MeV, the ground-state binding energy is*

$$|E_1| \;=\; \frac{(1/137.036)^2 \times 0.5100\ \mathrm{MeV}}{2} \;=\; 13.58\ \mathrm{eV}\,,$$

which is 0.20% *below the measured* 13.606 *eV—the same relative error as* m_e *itself, inherited rather than independent. If the electron mass formula is refined, the hydrogen spectrum follows automatically.*

6 The Vacuum Speaks

Our hydrogen atom sits in its ground state: silent, invisible, inert. Then a photon arrives, and the atom discovers that the vacuum is not empty.

The Lamb shift—the 1058 MHz splitting of the $2S_{1/2}$ and $2P_{1/2}$ states—scales as α^5: five powers of the bridge constant's imprint. Each power of α represents one more layer of interaction between the discrete (the atom's energy levels) and the continuous (the quantum vacuum). The Lamb shift is the discrete-to-continuous bridge made measurable.

The hyperfine transition—1420.405 MHz, the 21-cm line—scales as $\alpha^4(m_e/m_p)$. The mass ratio introduces the strong force (since m_p is set by QCD running under $b_3 = 7$), so the hyperfine line is a meeting point of electromagnetic and strong interactions, both encoded in the master quadratic.

Every stage of the atom's existence exhibits the same architecture:

Event	**Discrete**	**Continuous**
Quark confinement	baryon number	QCD flux tubes
Recombination	electron capture	photon field
Absorption/emission	energy levels E_n	electromagnetic field
Lamb shift	atomic state	vacuum fluctuations
Hyperfine transition	spin states	radio-frequency field
Stellar fusion	nuclear binding	thermal plasma
Proton decay	baryon number $\to 0$	radiation field

At every transition, the bridge between the discrete side and the continuous side is controlled by coupling constants determined by ϖ, G^*, and the integers $\{3, 4, 7, 13\}$.

The same structural motif appears in pure mathematics. The identity

$$\sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2} = \frac{2 + 4\lambda \coth(4\lambda) + 16\lambda^2 \operatorname{csch}^2(4\lambda)}{4}$$

where $\lambda = \varpi^2/G^{*2} = \pi/4$, passes from a discrete sum (left) to a closed-form expression involving the ontic ratio (right) through the Eisenstein series of the Gaussian lattice $\mathbb{Z}[i]$. The analogy is one of form, not mechanism: the Rydberg series arises from the Coulomb Green's function, the sum above from the Epstein zeta function. But the architectural crossing—from Σ to $\int$, from discrete to continuous—is governed by the same constant.

The bridge is not π. The bridge is G^*.

7 Bell's Theorem and the Gauss Constraint

Two hydrogen atoms, created together in a singlet state, are separated and measured along independently chosen axes. The correlations between outcomes violate Bell's inequality. No local hidden variable theory can reproduce them. How does a deterministic lattice produce nonclassical correlations?

Theorem 26 (Gauss Constraint Reduces DOF)**.** [Theorem] *The Gauss constraint from the lattice action imposes $\nabla \cdot \mathbf{J} = \rho$. This is one scalar constraint on three flux components, reducing the physical degrees of freedom from 3 to 2. The physical flux lives in the 2D transverse plane perpendicular to the propagation direction:*

$$\mathbf{J}_{\text{phys}}(\mathbf{k}) = \mathbf{J}(\mathbf{k}) - \hat{\mathbf{k}}\big(\hat{\mathbf{k}} \cdot \mathbf{J}(\mathbf{k})\big).$$

Theorem 27 (Classical Correlation from 3D Flux)**.** [THEOREM] *Without the Gauss constraint, the full three-component flux on S^2 yields the triangle correlation function*

$$E(\theta) \ = \ -\left(1 - \frac{2|\theta|}{\pi}\right),$$

which gives $S = 2$ for the CHSH parameter—exactly the Bell bound.

Theorem 28 (Quantum Correlation from 2D Flux)**.** [THEOREM + SELECTION] *After the Gauss projection, the transverse flux lives in a 2D plane admitting complexification $\psi = J_x + i\, J_y$* [SELECTION]*. The singlet correlation function for the resulting complex amplitude is*

$$E(\theta) \ = \ -\cos\theta\,,$$

which saturates the Tsirelson bound $S = 2\sqrt{2}$ for the CHSH inequality.

Proof sketch. In 2D, the uniform distribution on S^1 has density $P(\phi) = 1/(2\pi)$. A detector aligned at angle θ measures $\mathrm{sgn}(\cos(\phi - \theta))$. The product correlation integrates to

$$E(\theta) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{sgn}(\cos\phi)\,\mathrm{sgn}(\cos(\phi-\theta))\,d\phi = -\cos\theta.$$

The CHSH parameter $S = E(\theta_1) - E(\theta_2) + E(\theta_3) + E(\theta_4)$ is maximized at $\theta_j = \{0, \pi/4, \pi/2, \pi/4\}$, giving $S = 2\sqrt{2}$. Numerically verified with 2×10^6 Monte Carlo samples (13/13 tests pass). □

Scholium 29. The Gauss constraint is *local*—it acts site-by-site on the lattice. Yet its statistical consequence is nonclassical. The elevation from $S = 2$ to $S = 2\sqrt{2}$ does not require nonlocality, superdeterminism, or retrocausality. It requires only that a physical constraint reduces three degrees of freedom to two.

Remark 30 (Self-Referential Closure). [SELECTION] Measurement in FTD is not epistemological but dynamical. The observer $\mathcal{O} \subset \mathbf{L}$ is itself a manifested structure on the lattice. It observes $\mathbf{L}\backslash\mathcal{O}$ through the shared flux substrate. The coupling term $\mathcal{L}_{\text{coupling}} = -g_c{\cdot}s{\cdot}(\nabla{\cdot}\mathbf{J})$ requires a manifested entity ($s \neq 0$) to trigger flux concentration. Without a manifested observer, the flux evolves linearly and no "measurement" occurs. Consciousness, in this reading, is the self-referential closure of the projection: the point where the shadow becomes aware that it is a shadow, and in so doing, participates in the casting of light. This is structural realism: what exists fundamentally is mathematical structure, and both "mind" and "matter" are aspects of it.

8 Gravity and the Arrow of Time

Our hydrogen atom drifts for hundreds of millions of years, part of a diffuse cloud. Then gravity gathers the cloud. It contracts, heats, and eventually the core temperature reaches $\sim 10^7\,\mathrm{K}$. The weakest force has accomplished what no other could: concentrating enough matter to ignite fusion.

Theorem 31 (Projection Loss)**.** [THEOREM] *The projection $\Pi : \mathcal{V} \to \mathbf{L} \times \{-1, 0, +1\}^{|\mathbf{L}|}$ from the infinite-dimensional space of smooth vector fields to the finite lattice is necessarily non-injective. The codomain has cardinality 3^{N^3}; the domain is infinite-dimensional. Information is destroyed at each update cycle.*

Proposition 32 (Arrow of Time)**.** [THEOREM] *Given irreversible projection loss at rate $\gamma > 0$, the entropy $S = -\sum_v \rho(v) \ln \rho(v)$ is monotonically non-decreasing. The arrow of time is the direction of increasing projection loss.*

Conjecture 33 (Gravity as Bandwidth Saturation)**.** [CONJECTURE] *Where manifested particles cluster densely, the lattice's finite update rate creates a local processing deficit. The effective metric experienced by flux waves propagating through this region is $g^{\text{eff}}_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x)$, with $h_{\mu\nu} \propto \rho_{\text{manifested}}(x)$, recovering linearized general relativity. The full Einstein equations $R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}$ follow via the Deser iterative bootstrap: demanding self-consistency (the gravitational field's own energy must appear as source) and invoking Lovelock's uniqueness theorem in $D = 4$ spacetime.*

Remark 34. Newton's constant in lattice units is $G_N = 1/(b_3 + N_c)^2 = 1/(7+3)^2 = 0.01$ [CONJECTURE]. The gravitational hierarchy $\alpha_G \sim \alpha^{20}$ encodes the vast separation between Planck and astrophysical scales. Stellar fusion is controlled by quantum tunneling through the Coulomb barrier, with Gamow energy $E_G = 2 m_r c^2 (\pi \alpha Z_1 Z_2)^2$. The value of α from the master quadratic falls in the narrow window compatible with stellar lifetimes of $\sim 10^{10}$ years.

9 Death and Dissolution

At $\sim 10^7\,\text{K}$, two protons fuse: one undergoes beta-plus decay (converting to a neutron via the weak force), and the result is a deuteron, a positron, and a neutrino. Our hydrogen atom's proton has become part of a deuterium nucleus. The atom, as a distinct entity, is gone.

The deuteron captures another proton to form helium-3; two helium-3 nuclei fuse to form helium-4. In more massive stars, the chain continues: helium to carbon, carbon to oxygen, up to iron-56 at the bottom of the valley of stability. Each step is a discrete nuclear transition in a continuous plasma, with rates set by coupling constants from the integers $\{3, 4, 7, 13\}$. The entire periodic table is a consequence of these four numbers.

If the star is massive enough, the iron core collapses, halted by neutron degeneracy pressure. The supernova disperses all elements heavier than helium back into the interstellar medium. The cycle is ready to begin again.

And if baryon number is not exactly conserved—if the proton eventually decays ($\tau_p > 10^{34}$ years [3])—then the last discrete object crosses the final bridge into the continuum of radiation. Photons have no rest mass, no charge, no baryon number. They are pure field, pure wave, pure continuity. The integer becomes the irrational. The lattice point dissolves into the plane.

$$\sum \longrightarrow \int$$

The ratio $\lambda = \varpi^2 / G^{*2}$ was there before the atom, is there during the atom, and will be there after the atom. The atom is a transient excitation. The constants are not made of atoms.

10 The Honest Ledger

Scholium 35. A framework that claims everything derives nothing. This section catalogs what FTD actually proves, what it inserts, and what it borrows.

Epistemic Classification of Claims in This Paper

Claim	Status	Section
Axiom Zero (voxel = state + position)	[Axiom]	1
Two-layer ontology (flux/state)	[Axiom]	1
I_4, ϖ, G^* from Γ-functions	[Theorem]	2
Watson integral $W_3 = \Gamma(1/4)^4/(4\pi^3)$	[Theorem]	2
$\pi = 4\varpi^2/G^{*2}$	[Theorem]	2
Master quadratic roots $x_\pm$	[Theorem]	3
$x_+ = \alpha^{-1}$ (1.26 ppm)	[Conjecture]	3
$D = 3$ uniqueness: $\lfloor x_-^{(D)} \rfloor = D$ iff $D = 3$	[Theorem]	4
Gauge groups from Moore decomposition	[Theorem]	4
Electron mass formula	[Selection]	5
Higgs quartic $\lambda = 3/23$	[Selection]	5
Higgs mass $m_H = 125.69\,\mathrm{GeV}$	[Selection]	5
Hydrogen spectrum from α, m_e	[Parametric Insertion]	5
Gauss constraint $\to$ 2D flux	[Theorem]	7
Bell violation $S = 2\sqrt{2}$	[Theorem + Selection]	7
Projection is non-injective	[Theorem]	8
Arrow of time from projection loss	[Theorem]	8
Einstein equations (Deser bootstrap)	[Conjecture]	8
Self-referential closure (observer)	[Selection]	7

The Five Selection Principles

The strongest results depend on five non-derivable choices:

SP1. **Why the lemniscatic curve?** Among all elliptic curves, $E : y^2 = x^3 - x$ with $j = 1728$ is selected for maximal $\mathbb{Z}_4$ symmetry compatible with $\mathbb{Z}^2$. But $j = 0$ has $|\,\mathrm{Aut}\,| = 6$. No proof excludes non-CM curves.

SP2. **Why a quadratic?** Minimal polynomial for two roots. But cubic or transcendental forms are not excluded.

SP3. **Why coefficient 16?** Six arithmetic routes converge on $|\,\mathrm{Aut}(E)|^2 = 16$. Strongly motivated; not uniquely forced.

SP4. **Why $\varepsilon = e^\pi - \pi - 20$?** The near-integer property $e^\pi \approx \pi + 20$ is a known numerical coincidence. The decomposition $20 = 7 + 13 = b_3 + N_{\mathrm{eff}}$ is suggestive but possibly post hoc.

SP5. **Why these correction coefficients?** The 4-term precision series uses rationals from $\{3, 4, 7, 13\}$ to close the 1.26 ppm gap to sub-ppt. With four free coefficients and a small expansion parameter, matching a 12-digit target is algebraically possible regardless of the underlying physics.

Honest Accounting

The FTD project contains approximately:

- ~ 30 genuine derivations (from G^* and integers alone)
- ~ 50 parametric insertions (FTD values in standard physics formulas)
- $\sim 50+$ external physics results adopted without derivation

This paper focuses on the genuine derivations. The parametric insertions (e.g., m_e from the hierarchy formula) and external physics (e.g., the pp chain, neutron degeneracy) are used as context, not as claimed results.

What FTD Does Not Explain

- The cosmological constant
- The absolute mass scale (requires m_P as input)
- Individual quark masses (only ratios)
- Why the axiom itself is realized in nature

11 Falsification

A framework that cannot be wrong is not physics. FTD is falsified by:

1. **Precision α measurement.** If CODATA measurements move away from $x_+ = 137.036\,17\ldots$ at better than 10 ppm, the master quadratic is falsified. Current measurements are at 0.15 ppb, already constraining.
2. **Fourth fermion generation.** Discovery of a fourth generation with standard gauge couplings falsifies $\lfloor x_- \rfloor = 3$.
3. **Failure of the Bell mechanism.** Demonstration that no ensemble averaging over local deterministic lattice states can yield $S > 2$ would falsify the substrate-to-aggregate transition.
4. **Observable Lorentz violation.** Detection of energy-dependent photon speed with the wrong sign (superluminal rather than subluminal) at high energies would contradict the lattice dispersion relation.
5. **The Higgs quartic.** The prediction $\lambda = 3/23 = 0.1304$ is testable. Current measurement: 0.129 ± 0.01. The High-Luminosity LHC will sharpen this constraint.
6. **Projection residuals.** A quantitative derivation showing that the projection residual $\mathcal{R}(v) = \mathbf{J}(v) - \mathbf{J}_\Pi(v)$ does *not* produce inertial behavior would falsify the mass-as-residual conjecture.

Coda: The Bridge Remains

A cubic lattice with ternary states and local causality, combined with the geometry of the lemniscate of Bernoulli, produces a master quadratic whose roots determine the fine-structure constant and the number of color charges. From these two numbers and four framework integers, the gauge groups, the Higgs quartic, the Bell violation bound, and the hydrogen spectrum follow—conditionally on five stated selection principles. Every stage of a hydrogen atom's existence, from quark confinement to spectral lines to stellar fusion to final dissolution, is governed by this single algebraic structure.

Whether this contact between geometry and physics reflects coincidence, selection, or necessity is not for the framework to decide. It is for experiment.

The bridge is not π. The bridge is G^*.
And the story of one hydrogen atom is the story of every discrete thing that ever existed in a continuous universe.

Foundational Ternary Dynamics: a framework, not a faith.

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